TP2/RR2 for SPY and SPX options

May 19, 2026

1 Summary statistics

The options data on SPY and SPX covers the period 1-Jan-25 to 29-Aug-25, for a total of 165 trading days.

The range of maturities of traded options is wider for SPY than for SPX options. Figure 1 shows the number of traded strikes/day vs maturity T for the SPY options. In contrast, the great majority of SPXW options have maturities up to 200 days. Figure 2 shows the number of traded strikes for SPXW calls/puts on 2-Jan-2025.

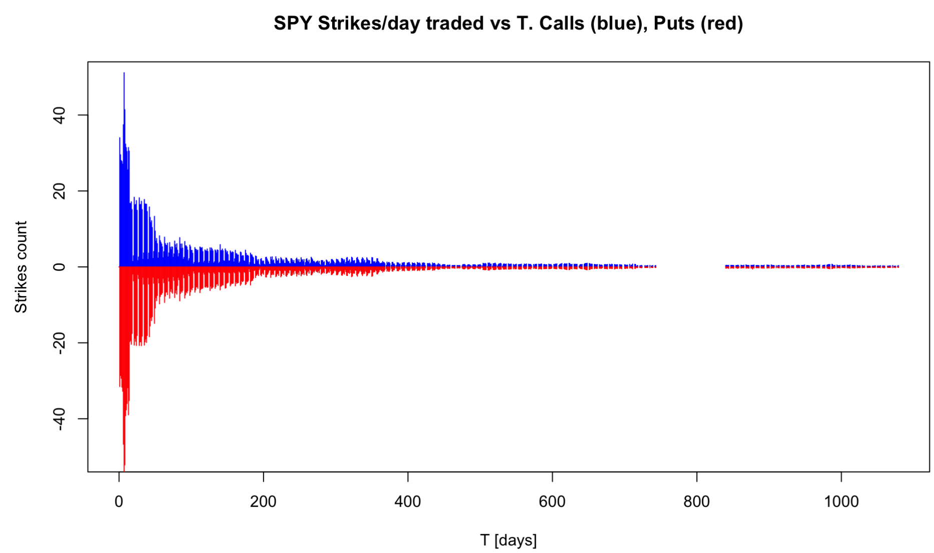


Figure 1: The number of strikes traded/day for SPY options aggregated over 2025, separately for calls and puts.

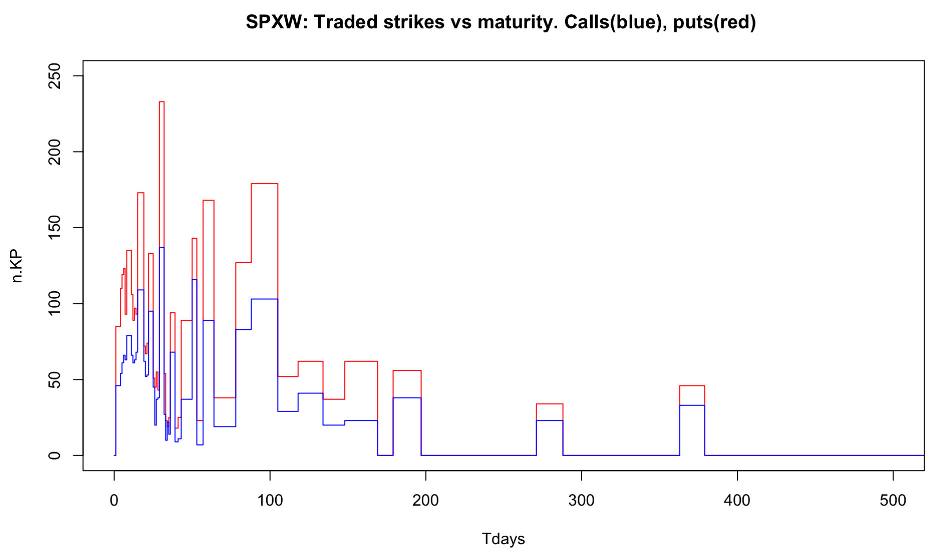


Figure 2: The number of strikes traded for SPXW options on 2-Jan-25 vs maturity T, separately for calls and puts.

2 Violation rates

The violation rates for both SPX and SPY are shown in Table 1. The TP_2/RR_2 violations are computed using mid-prices. The result is consistent with Mike's Figure 22 which shows larger violations for SPX puts than calls. The difference between the violation rates for calls and puts becomes larger when restricting only to PM-settled options SPXW.

Restricting to OTM options. Table 2 compares the violation ratios for all strikes with those obtained by restricting only to OTM strikes. For simplicity I used the spot price S_0 on the pricing date to distinguish OTM from ITM options. These results include both AM- and PM-settled options.

Theoretical expectation in the BS model with positive drift $r - q \geq 0$ is that TP_2 holds for European calls with all strikes, while RR_2 holds only for European OTM puts. No such restrictions are expected for American options.

Thus, for SPX we expect to see comparable TP2 violation rates for OTM and for all-K, and higher RR2 violations for all-K compared to OTM puts alone. The numbers show higher TP2 violation rates for OTM calls, and slightly larger RR2 violation rates for puts (opposite to expectations).

For SPY we expect to see similar violation rates for OTM/all-K. In fact we observe higher violation rates for OTM calls

3 Correlation SPY-SPX violation rates

Figure 1 shows the daily violation rates, comparing SPY and SPX, separately for calls and puts. The horizontal lines show the average violation rates.

Table 1: Violation rates for SPY and SPX options. First two rows include both AM- and PM-settled SP500 options. The rows 3,4 show the results keeping only PM-settled options (SPXW).

	n_{tot}	$[n_{vio}]_{mid}$	vr_{mid}	$[n_{vio}]_{bid-ask}$	$vr_{bid-ask}$
SPX calls	167,812,336	1,154,833	0.688%	140,591	0.084%
SPX puts	429,757,716	10,955,227	2.550%	1,839,288	0.428%
SPXW calls	63,081,514	177,028	0.281%	3,899	0.006%
SPXW puts	144,042,772	3,329,029	2.311%	407,911	0.283%
SPY calls	43,376,334	176,859	0.408%	10,909	0.025%
SPY puts	62,305,648	1,182,236	1.897%	232,228	0.373%

Table 2: Violation rates for SPY and SPX options, comparing with their restriction to OTM strikes. Midprices are used for all cases. SPX includes both AM- and PM-settled options.

	vr_{all-K}	vr_{OTM}
SPX + SPXW calls	0.688%	1.068%
SPX + SPXW puts	2.550%	2.606%
SPY calls	0.408%	1.080%
SPY puts	1.897%	1.960%

Days of high violation rates for SPX coincide mostly with those of high violation rates for SPY. The correlations are similar to those obtained by Mike.

Table 3: Correlations among daily violation rates for SPY and SPX. The first two rows include all SPX options, and the last two rows keep only PM-settled options SPXW.

	Pearson	Spearman
Calls SPY-SPX(all)	71.1%	53.6%
Puts SPY-SPX(all)	88.5%	85.9%
Calls SPY - SPXW	69.4%	86.2%
Puts SPY - SPXW	86.2%	69.4%

The same data is shown as scatter plots in Figure 2 which shows better the correlation. The lines show a linear fit to the daily violation rates.

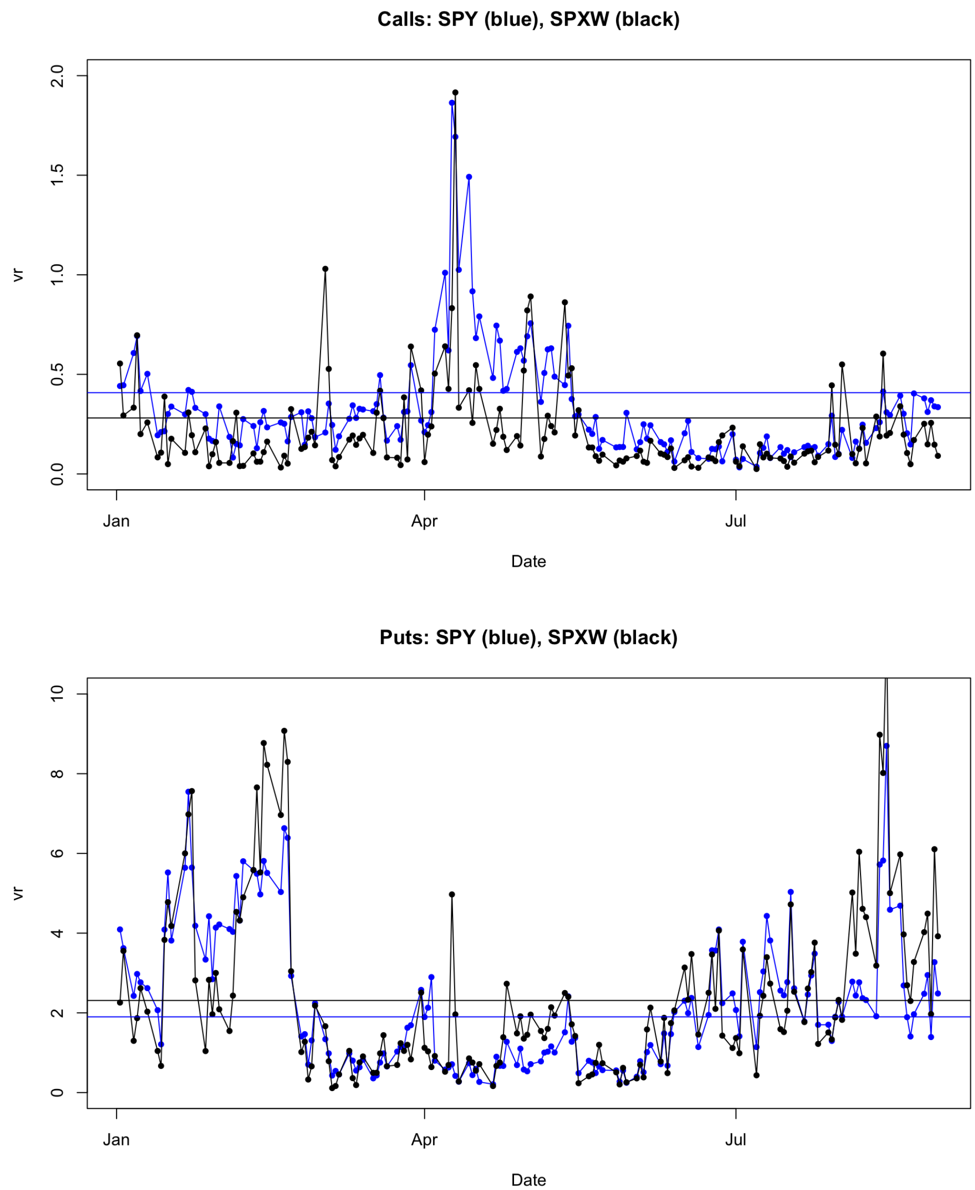
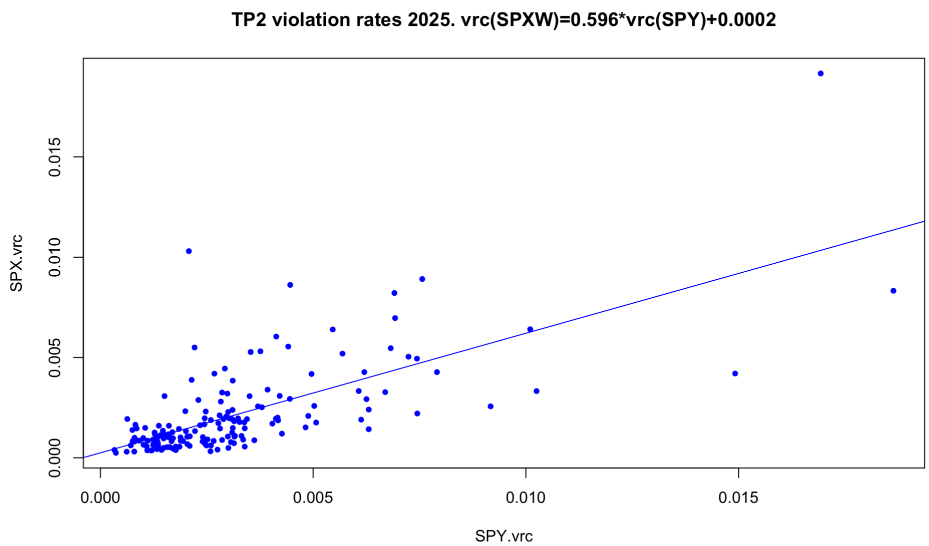


Figure 3: The violation rates for TP2/RR2 violations for each day in the 2025 dataset, comparing SPY and SPXW (PM-settle only). The horizonal lines show the average violation rates from Table 1.



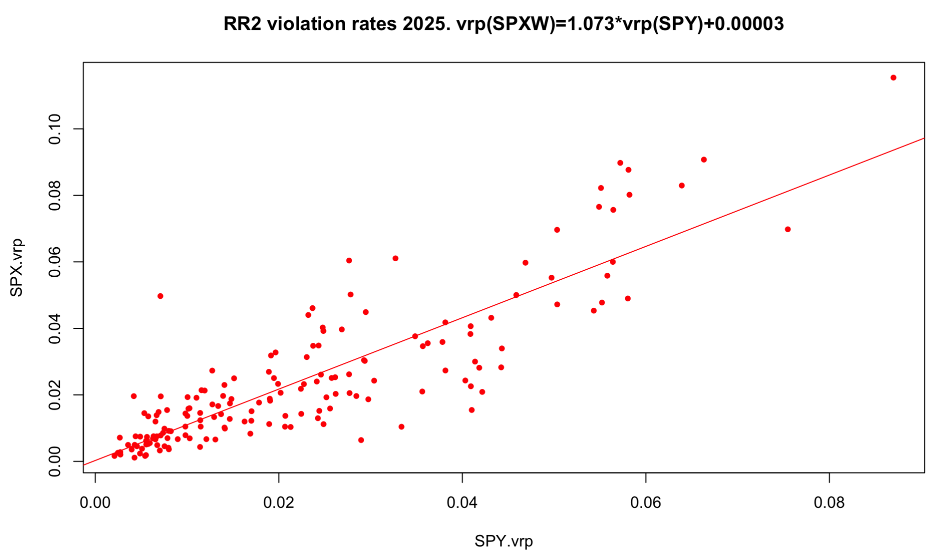


Figure 4: Scatter plots for the daily violation rates for SPXW and SPY, which show the correlation among them.